1 Basic tests

Theorem 1.1. *Case 1: Environment ends with simple text.* ♡

Theorem 1.2. *Case 2: Environment ends with display*

$$1 + 1 = 2. \qquad \heartsuit$$

Theorem 1.3. *Case 3: Environment ends with equationlike (numbered)*

$$1 + 1 = 2. \tag{1}$$ ♡

Theorem 1.4. *Case 4: Environment ends with*

- *a*
- *list.* ♡

2 Other simple tests

These test multiple or nested boxes inside of theoremlike environments.

Remark 2.1. Case 1: Environment ends with simple text. But first there is an equation

$$1 + 1 = 2 \tag{2}$$

and then the simple text. ◇

Remark 2.2. Case 2: Environment ends with display

1. but first
2. a list

$$1 + 1 = 2. \qquad \diamond$$

Remark 2.3. Case 3: Environment ends with align following

- a
- list

and

`display math`

`again`

$$x = 2 \tag{3}$$

$$y = 3. \quad \text{this is a very wide box that uses many words to fill the line} \tag{4}$$ ◇

Remark 2.4. Case 4: Environment ends with

- a
- list.
- that is nested
    1. inside another
    2. list ◇

Note 2.1. Case 4: Environment ends with

- a
- list
- with a display
$$a^2 + b^2 = c^2$$
    1. and there is
    2. a nested list
- but the outer list has
- the final item •

Note 2.2.

- a
- list
    1. and there is
    2. a nested list
    3. with a display
$$a^2 + b^2 = c^2 \quad \bullet$$

Note 2.3.

- a
- list
    1. and there is
    2. a nested list
    3. with align

$$a^2 + b^2 = c^2 \tag{5}$$
$$3^2 + 4^2 = 5^2 \tag{6}$$
•

Note 2.4.

- a
- list
  1. and there is
  2. a nested list
  3. with align

$$a^2 + b^2 = c^2 \tag{7}$$
$$3^2 + 4^2 = 5^2 \tag{8}$$

Then there is some text at the end •

Note 2.5. First an align

$$a^2 + b^2 = c^2 \tag{9}$$
$$3^2 + 4^2 = 5^2 \tag{10}$$

Then some text,

1. and then
2. a list. •

3 some other messy tests

Theorem 3.1. *regular text* ♡

Theorem 3.2. *This ends in a display*

$$3^2 + 4^2 = 5^2. \qquad \heartsuit$$

Theorem 3.3. *This ends in a display*

$$3^2 + 4^2 = 5^2$$

more text

$$a = b \tag{11}$$
$$c = d \tag{12}$$
♡

Theorem 3.4. *This ends in a display*

$$3^2 + 4^2 = 5^2.$$

more text

$$a = b$$

abc ♡

Theorem 3.5. *This ends with*

1. *an*
2. *enumerated*
3. *list.* ♡

Theorem 3.6. *This ends with*

1. *an*
2. *enumerated*
3. *list*
4. *ending in equation:*

$$a = b \qquad \heartsuit$$

Theorem 3.7. *This ends with*

1. *an*
2. *enumerated*
3. *list.*
   (a) *abc*
4. *blub* ♡

Theorem 3.8. *This ends with*

1. *an*
2. *enumerated*
3. *list.*
   (a) *abc* ♡

3.1 tests of nested theoremlike environments

Theorem 3.9. *regular text*

Lemma 3.1. *blub; lemma is not endmarked*

Lemma 3.2. *abc*

abc

1. *abc*
   - *blub*

2. *jhdgd* $\heartsuit$

Theorem 3.10. *regular text*

Remark 3.1. blub; remark *is* endmarked $\diamond$

$\heartsuit$

Theorem 3.11. *regular text*

Remark 3.2. blub; remark *is* endmarked $\diamond$

Lemma 3.3. *abc; lemma is not endmarked*

$$abc$$

1. *abc*
    - *blub*
2. *jhdgd* $\heartsuit$

3.2 proof tests

Proof. regular text □

Proof. This ends in a display

$$3^2 + 4^2 = 5^2. \qquad \square$$

Proof. This ends in a display

$$3^2 + 4^2 = 5^2.$$

more text

$$a = b \tag{13}$$
$$c = d \tag{14}$$
□

Proof. This ends in a display

$$3^2 + 4^2 = 5^2.$$

more text

$$a = b$$

abc □

Proof. This ends with

1. an
2. enumerated
3. list. □

Proof. This ends with

1. an
2. enumerated
3. list
4. ending in equation:

$$a = b \qquad \square$$

Proof. This ends with

1. an
2. enumerated

3. list.

    (a) abc

4. blub □

Proof. This ends with

1. an
2. enumerated
3. list.

    (a) abc □

4 Comprehensive test of endmarkable environments

[in progress] This section aims to have one simple example for every possible environment that could be endmarked, so that we can confirm support.

Subsections are intended to group environment types according to how they are handled internally; revise as necessary.

Many of the examples and descriptions are from the amsmath documentation: Section 3, Displayed equations.

4.1 equationlike

These environments are full line width. They do not use horizontal alignment markers (&).

Test 4.1. `equation`: a single equation with an automatically generated number

$$\int_a^b f(x)\,dx \tag{15} \qquad \diamond$$

Test 4.2. `equation*`: unnumbered variant

$$\int_a^b f(x)\,dx \qquad \diamond$$

Test 4.3. `[..]`: The classic display math. Internally, amsmath redefines this as equation*

$$\int_a^b f(x)\,dx \qquad \diamond$$

Test 4.4. `$$..$$`: People still use this, but maybe we don't want to support it. By default, it doesn't work well with `qedhere` (end mark not pushed to end of line).

$$\int_a^b f(x)\,dx$$

$\diamond$

Test 4.5. `multline`: a variation of equation, for long equations with linebreaks. The first line is aligned left, middle lines are centered, last line is aligned right. Equation number is on the first line.

$$\begin{multline} a+b+c+d+e+f+a+b+c+d+e+f \tag{16} \\ +g+h+g+h+g+h+ \\ +i+j+k+l+m+n+i+j+k+l+m+n \quad \diamond \end{multline}$$

Test 4.6. `multline*`: unnumbered variant

$$a+b+c+d+e+f+a+b+c+d+e+f \\ +g+h+g+h+g+h+ \\ +i+j+k+l+m+n+i+j+k+l+m+n \quad \diamond$$

Test 4.7. `gather`: a group of numbered equations with no horizontal alignment; each line is horizontally centered.

$$\int_a^b f(x)\,dx \tag{17}$$

$$\int_a^b f(x)g(x)h(x)\,dx \tag{18}$$

$\diamond$

Note: currently `gather` is handled the same way as `align`. It's unclear (to me, NGJ) whether it should be grouped with the other "alignlike" environments for that reason.

Test 4.8. `gather*`: unnumbered variant

$$\int_a^b f(x)\,dx$$

$$\int_a^b f(x)g(x)h(x)\,dx \quad \diamond$$

Test 4.9. `displaymath`: This is an older environment, not one of the amsmath ones; the ams doc says it's functionally equivalent to equation*

$$\int_a^b f(x)\,dx \quad \diamond$$

4.2 alignlike

These environments are full line width. They use horizontal alignment markers (&).

Test 4.10. `align`: for two or more equations; this test has just one alignment column

$$\int_a^b f(x)\,dx = 1 \tag{19}$$

$$\int_a^b g(x)+h(x)\,dx = a+b+c \tag{20}$$

$\diamond$

Test 4.11. `align`: another test, with multiple alignment columns

$$\begin{align}\int_a^b f(x)\,dx &= 1 & x &= y+z & p &= 0 \tag{21}\\ \int_a^b g(x)+h(x)\,dx &= a+b+c & x'+y' &= z' & q &= 1 \tag{22}\end{align}$$ ◇

Test 4.12. `align*`: unnumbered variant; one column

$$\begin{align*}\int_a^b f(x)\,dx &= 1\\ \int_a^b g(x)+h(x)\,dx &= a+b+c\end{align*}$$ ◇

Test 4.13. `align*`: unnumbered variant; multiple columns

$$\begin{align*}\int_a^b f(x)\,dx &= 1 & x &= y+z & p &= 0\\ \int_a^b g(x)+h(x)\,dx &= a+b+c & x'+y' &= z' & q &= 1\end{align*}$$ ◇

Test 4.14. `alignat`: allows the horizontal space between equations to be explicitly specified; takes an argument for number of "equation columns".

$$\begin{alignat}{2}x &= y_1 - y_2 + y_3 - y_5 + y_8 - \dots &\quad& \text{by (C)} \tag{23}\\ &= y' \circ y^* && \text{by (D)} \tag{24}\\ &= y(0)y' && \text{by Axiom 1.} \tag{25}\end{alignat}$$ ◇

Test 4.15. `alignat*`: unnumbered variant

$$\begin{alignat*}{2}x &= y_1 - y_2 + y_3 - y_5 + y_8 - \dots &\quad& \text{by (C)}\\ &= y' \circ y^* && \text{by (D)}\\ &= y(0)y' && \text{by Axiom 1.}\end{alignat*}$$ ◇

Test 4.16. `flalign`: full length alignment; stretches to fill line, leaving space for equation numbers and endmark (if present)

$$\begin{flalign}x &= y & X &= Y \tag{26}\\ x' &= y' & X' &= Y' \tag{27}\\ x+x' &= y+y' & X+X' &= Y+Y' \tag{28}\end{flalign}$$ ◇

Test 4.17. `flalign*`: unnumbered variant

$$\begin{flalign*}x &= y & X &= Y\\ x' &= y' & X' &= Y'\\ x+x' &= y+y' & X+X' &= Y+Y'\end{flalign*}$$ ◇

4.3 inner math environments

These environments go inside other displayed math environments.
Note: Maybe these should all be moved to the "unsupported" section, and their endmarks can be handled by the surrounding environments.

Test 4.18. `split`: for a *single* equation split among multiple lines, like `multiline`, but with some alignment points.
This `split` is inside `equation`.

$$\begin{split} H_c &= \frac{1}{2n} \sum_{l=0}^{n} (-1)^l (n-l)^{p-2} \sum_{l_1+\dots+l_p=l} \prod_{i=1}^{p} \binom{n_i}{l_i} \\ &\quad \cdot [(n-l)-(n_i-l_i)]^{n_i-l_i} \cdot \Bigl[(n-l)^2 - \sum_{j=1}^{p} (n_i-l_i)^2 \Bigr]. \end{split} \tag{29}$$

Test 4.19. `split`: This `split` is inside `gather`, between other lines

$$a+b=c \tag{30}$$

$$\begin{split} H_c &= \frac{1}{2n} \sum_{l=0}^{n} (-1)^l (n-l)^{p-2} \sum_{l_1+\dots+l_p=l} \prod_{i=1}^{p} \binom{n_i}{l_i} \\ &\quad \cdot [(n-l)-(n_i-l_i)]^{n_i-l_i} \cdot \Bigl[(n-l)^2 - \sum_{j=1}^{p} (n_i-l_i)^2 \Bigr]. \end{split} \tag{31}$$

$$\int_a^b f(x)\,dx \tag{32}$$

◇

The width of the following environments is just the width of the content. Therefore, maybe they should *not* be endmarkable? (I'm not sure.) Note: there are options to set their vertical alignment; maybe that is useful?

Each of these is inside `equation`. Each takes an optional argument for vertical positioning (baseline); demonstrated just with the first one.

Test 4.20. `gathered`: [c] (the default)

$$\begin{gathered} \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ a+b+c+d=e+f+g+h+i+j+k \end{gathered} \tag{33}$$

◇

Test 4.21. `gathered`: [t] (baseline top)

$$\begin{gathered}[t] \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ a+b+c+d=e+f+g+h+i+j+k \end{gathered} \tag{34}$$ ◇

Test 4.22. `gathered`: [b] (baseline bottom)

$$\begin{gathered}[b] \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ a+b+c+d=e+f+g+h+i+j+k \end{gathered} \tag{35}$$ ◇

Test 4.23. `aligned`:

$$\begin{aligned} \int_a^b f(x)\,dx &= 2 \\ \int_a^b f(x)\,dx &= x+y+z \\ \int_a^b f(x)\,dx &= 1 \\ a+b+c+d &= e+f+g+h+i+j+k \end{aligned} \tag{36}$$ ◇

Test 4.24. `alignedat`:

$$\begin{alignedat}{2} x &= y_1 - y_2 + y_3 - y_5 + y_8 - \dots &\quad& \text{by (C)} \\ &= y' \circ y^* && \text{by (D)} \\ &= y(0) y' && \text{by Axiom 1.} \end{alignedat} \tag{37}$$ ◇

Test 4.25. `cases`:

$$P_{r-j} = \begin{cases} 0 & \text{if $r-j$ is odd,} \\ r!\,(-1)^{(r-j)/2} & \text{if $r-j$ is even.} \end{cases}$$ ◇

Test 4.26. `dcases`: This is a variant that the amsdocs mention, but is provided by the `mathtools` package. It makes the case lines displaystyle instead of (the default) textstyle. So, maybe we don't want to support this directly (especially not if we decide not to make `cases` endmarkable in the first place).
[*no example here*] ◇

4.4 lists

Test 4.27. `enumerate`: items are counted

1. top
2. middle
3. bottom ◇

Test 4.28. `itemize`: items are not counted

- top
- middle
- bottom ◇

Test 4.29. `description`: items have labels

The first one: goes here

Another: and this is the item content

The final one: these are often used for defintions ◇

Test 4.30. `list`: This is the basic environment that others are derived from. Adding thmmark support for this might either (a) automatically give it to other lists or (b) make us confused because other environments use `list` in subtle ways (e.g., trivlists).
Update: I've added this, and it's fine; it doesn't change the others. Furthermore, the hooks still have to be added for `itemize`, `enumerate`, and `description`; it's not enough to just set the hooks for `list`.

first

middle

last ◇

4.5 text blocks

(todo: quote, verbatim, other trivlist?)

Test 4.31. `quote`:

> A quotation goes here. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. ◇

verbatim

Currently this only sort of works. The `\end{verbatim}` must be separated by a space, but not a newline, from the contents. This could have to do with the `hmode` v.s. `vmode` checks in the thmmark code.

Test 4.32. `verbatim`: Note that linebreaks are also treated verbatim.

```
Some verbatim symbols \ & $ etc.
More verbatim symbols \ & $ etc. More verbatim symbols \ & $ etc. More verbatim symbols \ &
(The previous line intentionally runs off the page)
```
◇

Test 4.33. `verbatim`: (Another test) If the last line is *just a little* too long, it will push the endmark into the margin.

```
A123456789.B123456789.C123456789.D123456789.E123456789.F123456789
```
◇

Test 4.34. `verbatim`: (Another test) If the last line is *a little more* too long, it will push the endmark down.

```
A123456789.B123456789.C123456789.D123456789.E123456789.F123456789.
```
◇

some other environments

I'm not sure whether these go with this group, or somewhere else.

Test 4.35. `center`: center the contents

some centered text, or equation, or other environment X

◇

Test 4.36. `tikzpicture`: often used for diagrams, but of course requires tikz [*no example here*] ◇

Test 4.37. `tikzcd`: another popular one, also requires extra packages [*no example here*] ◇

4.6 tablelike

See source2e: File 41 ltab.dtx: Tabbing, Tabular, and Array Environments.
Explanation and examples from `https://latexref.xyz`.

Test 4.38. `tabbing`: manually set and use tab stops

row1-col1	row1-col2	this one has a longer heading	r1-c4
row2-col1	row2-col3	.	.
r3-c1	next field	a	b

◇

Test 4.39. `tabular`: a table

Player name	*Career home runs*
Hank Aaron	755
Babe Ruth	714
Hank Aaron (repeat)	755
Babe Ruth (repeat)	714

◇

4.7 unsupported

These environments are intentionally unsupported. The marks may or may not be placed as expected, depending on the surrounding environment(s).

Test 4.40. `eqnarray`: not recommended, for many reasons documented in Avoid `eqnarray`! (TUGboat 33:1, 2012)

$$\int_a^b f(x)\,dx \quad = 1. \tag{38}$$

◇

Test 4.41. `pmatrix`: other matrix environments are likewise not (directly) supported: `smallmatrix`, `bmatrix`, `Bmatrix`, `vmatrix`, and `Vmatrix`. Just let the surrounding environment handle the end mark.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

◇

Test 4.42. `array`: works inside displayed math; this one is in equation*. Just let the surrounding environment handle the end mark.

$$\chi(x) = \left| \begin{array}{ccc} x-a & -b & -c \\ -d & x-e & -f \\ -g & -h & x-i \end{array} \right|$$

◇

5 Other Tests

Here are tests of two things:

1. a command to disable the thmmark for one box
2. a command to put the thmmark on a previous line

Test 5.1. `\noqed`

$$\chi(x) = \left| \begin{array}{ccc} x-a & -b & -c \\ -d & x-e & -f \\ -g & -h & x-i \end{array} \right|$$

Test 5.2. `\noqed` followed by `\qedprev`
Combining these could be a way to manually mark the bottom of a box.
Note: an `\hfill` is required here.

$$\chi(x) = \begin{vmatrix} x-a & -b & -c \\ -d & x-e & -f \\ -g & -h & x-i \end{vmatrix} \qquad \diamond$$

Paragraph after environment, to check spacing. Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus.

Test 5.3. (duplicate, with no extra commands, to check spacing)

$$\chi(x) = \begin{vmatrix} x-a & -b & -c \\ -d & x-e & -f \\ -g & -h & x-i \end{vmatrix} \qquad \diamond$$

Paragraph after environment, to check spacing. Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus.

Test 5.4. `\noqed` does not work here

$$x = 2 \tag{39}$$
$$y = 3 \tag{40}$$
$$P_{r-j} = \begin{cases} 0 & \text{if } r-j \text{ is odd,} \\ r!\,(-1)^{(r-j)/2} & \text{if } r-j \text{ is even} \\ \text{other} & \text{case, for testing.} \end{cases} \qquad \diamond \tag{41}$$

Test 5.5. `\noqed` does not work here

$$x = 2 \tag{42}$$
$$y = 3 \qquad \diamond \tag{43}$$

6 align and friends with tall lines

Test 6.1. `align`: with `cases`

$$\text{words to push this content to the right edge} \qquad x = 2 \tag{44}$$

$$y = \begin{cases} \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \end{cases} \tag{45}$$

◇

Test 6.2. Similar to above, but manually reposition endmark with

```
\par\vspace{-1.6\baselineskip}~
```

after `\end{align}`

$$\text{words to push this content to the right edge} \qquad x = 2 \tag{46}$$

$$y = \begin{cases} \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \end{cases} \tag{47}$$

◇